

Technical Note No. 6*
Options, Futures, and Other Derivatives
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**Differential Equation for Price of a Derivative
on a Stock Providing a Known Dividend Yield**

Define f as the price of a derivative dependent on a stock that provides a dividend yield at rate q . We suppose that the stock price, S , follows the process

$$dS = \mu S dt + \sigma S dz$$

where dz is a Wiener process. The variables μ and σ are the expected growth rate in the stock price and the volatility of the stock price, respectively. Because the stock price provides a dividend yield, μ is only part of the expected return on the stock.¹

Because f is a function of S and t , it follows from Ito's lemma that

$$df = \left(\frac{\partial f}{\partial S} \mu S + \frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 \right) dt + \frac{\partial f}{\partial S} \sigma S dz$$

Similarly to the procedure described in the text for a non-dividend paying stock, we can set up a portfolio consisting of

$$\begin{array}{ll} -1 : & \text{derivative} \\ + \frac{\partial f}{\partial S} : & \text{stock} \end{array}$$

If Π is the value of the portfolio,

$$\Pi = -f + \frac{\partial f}{\partial S} S \tag{1}$$

and the change, $\Delta\Pi$, in the value of the portfolio in a time period Δt is as given by:

$$\Delta\Pi = \left(-\frac{\partial f}{\partial t} - \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 \right) \Delta t$$

In time Δt the holder of the portfolio earns capital gains equal to $\Delta\Pi$ and dividends on the stock position equal to

$$qS \frac{\partial f}{\partial S} \Delta t$$

Define ΔW as the change in the wealth of the portfolio holder in time Δt . It follows that

$$\Delta W = \left(-\frac{\partial f}{\partial t} - \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 + qS \frac{\partial f}{\partial S} \right) \Delta t \tag{2}$$

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¹ In a risk-neutral world $\mu = r - q$ as indicated in the text.

Because this expression is independent of the Wiener process, the portfolio is instantaneously riskless. Hence

$$\Delta W = r\Pi \Delta t \tag{3}$$

Substituting from equations (1) and (2) into equation (3) gives

$$\left(-\frac{\partial f}{\partial t} - \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 + qS \frac{\partial f}{\partial S}\right) \Delta t = r \left(-f + \frac{\partial f}{\partial S} S\right) \Delta t$$

so that

$$\frac{\partial f}{\partial t} + (r - q)S \frac{\partial f}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf$$

This is the equation in the text.